

Ay 7 Inferencia

Estimación de Momentos: Sea una muestra iid $X_1, \dots, X_n$

σ dimensión K

$$E(X^j) = \mu_j(\theta) \quad \wedge \quad m_j = \frac{1}{n} \sum_{i=1}^n X_i^j \quad j=1..K$$

Formamos K ecuaciones

$$m_j = \mu_j(\theta) \quad j=1..K$$

Las K soluciones son los estimadores $\hat{\theta}_1, \dots, \hat{\theta}_K \rightarrow$ No son únicos se pueden obtener diversos Estimadores de momentos

EMV:

$$L(\theta|x) = \prod_{i=1}^n f(x_i|\theta) \quad , \quad L(\theta_1|x) > L(\theta_2|x)$$

$$\uparrow$$

$$P_{\theta_1}(X=x) > P_{\theta_2}(X=x)$$

es más plausible ver la muestra si el parámetro es θ_1

Por lo tanto

$$\hat{\theta}_{EMV} = \arg \max_{\theta \in \Theta} [L(\theta)] = \arg \max_{\theta \in \Theta} \left[\underbrace{\log L(\theta)}_{l(\theta)} \right]$$

1) Definimos $l(\theta)$

2) $\frac{\partial l(\theta)}{\partial \theta_j} = 0 \quad j=1..K \rightarrow$ Encontrar θ que cumple la ecuación y es el $\hat{\theta}_{EMV}$

3) $\frac{\partial^2 l(\hat{\theta}_{EMV})}{\partial \theta_j^2} < 0 \rightarrow$ asegurarnos que es un máximo

*Teorema: Sea una muestra $X_1, \dots, X_n \sim f(x|\theta)$, T suficiente para θ .

Entonces la función de verosimilitud $L(\theta)$ depende de $X_1, \dots, X_n$ solo a través del estadístico suficiente. Luego el EMV de θ depende de los datos solo a través del estadístico suficiente $T(X)$.

*Teorema: Principio de invariancia del EMV

Si $\hat{\theta}_{EMV} \rightarrow$ Para cualquier función $g(\theta)$ el emv de $g(\theta)$ es $g(\hat{\theta}_{EMV})$

ej/ $X_1, \dots, X_n \sim f(x|\lambda) = \begin{cases} \lambda x^{\lambda-1} & 0 \leq x \leq 1 \\ 0 & \text{e.o.c} \end{cases}$

EMV/ $m = \bar{X}$

$$E(X) = \int_0^1 x \lambda x^{\lambda-1} dx = \frac{\lambda}{\lambda+1} x^{\lambda+1} \Big|_0^1 = \frac{\lambda}{\lambda+1}$$

$m = E(X)$

$$\bar{X} = \frac{\lambda}{\lambda+1} \Rightarrow \lambda(\bar{X}-1) + \bar{X} = 0$$

$$\hat{\lambda}_{EMV} = \frac{\bar{X}}{1-\bar{X}}$$

EMV/ $L(\lambda|x) = \lambda^n (\prod x_i)^{\lambda-1}$

$$l(\lambda) = n \log(\lambda) + (\lambda-1) \sum \log(x_i)$$

$$\frac{\partial l(\lambda)}{\partial \lambda} = \frac{n}{\lambda} - \sum \log(x_i) = 0 \rightarrow \hat{\lambda}_{EMV} = \frac{-\sum \log(x_i)}{n}$$

$$\frac{\partial^2 l(\lambda)}{\partial \lambda^2} = -\frac{n}{\lambda^2} < 0 \rightarrow \text{máximo}$$

c) media

$$P_r(X \leq c) = 1/2$$

$$\int_0^c \lambda x^{\lambda-1} dx = 1/2$$

$$c^\lambda = 1/2 \quad / \lambda$$

$$\lambda l_0(c) = -l_0(2)$$

$$l_0(c) = l_0(2^{-1/\lambda}) / \exp$$

$$c = 2^{-1/\lambda} \rightarrow \text{EMV de la mod. } \hat{c}_{\text{EMV}} = 2^{-1/\hat{\lambda}_{\text{EMV}}}$$

P2) $m_1 = \bar{X}$
 $m_2 = \bar{X}^2$

$$E(X) = 2 \cdot (-1) + B \cdot 0 + (1-\alpha-B) \cdot 1$$

$$= 1-2\alpha-B$$

$$E(X^2) = 2 \cdot (-1)^2 + B \cdot 0^2 + (1-\alpha-B) \cdot 1^2$$

$$E(X^3) = 1-B$$

$$\hat{\alpha} = \frac{1-B-\bar{X}}{2} = \frac{\bar{X}^2 - \bar{X}}{2}$$

$$\bar{X} = 1-2\alpha-B$$

$$\bar{X}^2 = 1-B \rightarrow \hat{B}_{\text{ML}} = 1-\bar{X}^2$$

EMV $\frac{\partial L}{\partial \alpha} = 0$
 $\frac{\partial L}{\partial \beta} = 0$

$$L(\theta|x) = 2^A \cdot B^C \cdot (1-\alpha-B)^D$$

$$l(\theta) = A l_0(\alpha) + C l_0(B) + D l_0(1-\alpha-B)$$

$$\frac{\partial l}{\partial \alpha} = \frac{A}{\alpha} - \frac{D}{1-\alpha-B} = 0 \rightarrow A - A\alpha - AB = 2D$$

$$B = \frac{A - A\alpha - 2D}{A}$$

$$\frac{\partial l}{\partial B} = \frac{C}{B} - \frac{D}{1-\alpha-B} = 0 \rightarrow C - C\alpha - CB = \beta D$$

$$C - C\alpha - C \frac{(A - A\alpha - 2D)}{A} = \frac{(A - A\alpha - 2D) \cdot D}{A}$$

$$AC - AC\alpha - CA + CA\alpha + CD\alpha = AD - AD\alpha - 2D^2$$

$$C\alpha = A - A\alpha - 2D$$

$$2(C+A+D) = A$$

$$\alpha = \frac{A}{2(C+A+D)}$$

$$B = \frac{A - \frac{A}{2} - \frac{AD}{2(C+A+D)}}{1} = 1 - \frac{A}{2(C+A+D)} = \frac{C}{2(C+A+D)}$$

c) $P_r(X=1)_{\text{ML}}$

$$= 1 - \frac{A}{n} - \frac{C}{n}$$

$$= \frac{n - A - C}{n}$$

$$= \frac{D}{n}$$

P3

$X \sim \text{Gamma}(\alpha, \theta)$

$\alpha, \theta > 0$

$$f(x|\alpha, \theta) = \frac{B^\alpha}{\Gamma(\alpha)} x^{\alpha-1} \exp(-Bx) \quad 0 < x < \infty$$

$$\mu_1 = \bar{X}$$

$$\mu_2 = \bar{X^2}$$

$$\begin{aligned} E(X) &= \int_0^\infty \frac{B^\alpha}{\Gamma(\alpha)} x^\alpha \exp\{-Bx\} \\ &= \frac{\Gamma(\alpha+1)}{B \Gamma(\alpha)} \underbrace{\left(\frac{B^{\alpha+1}}{\Gamma(\alpha+1)} x^{(\alpha+1)-1} \exp\{-Bx\} \right)}_1 \end{aligned}$$

$$E(X) = \frac{\alpha}{B}$$

$$\bar{X} = \frac{\alpha}{B}$$

$$\bar{X^2} = \frac{\alpha^2}{B^2} + \frac{\alpha}{B}$$

$$B\bar{X} = \alpha$$

$$\Rightarrow \bar{X} = \frac{B^2 \bar{X^2}}{\alpha^2} + \frac{B \bar{X}}{\alpha}$$

$$\bar{X^2} - \bar{X}^2 = \frac{\bar{X}}{B}$$

$$\hat{B}_m = \frac{\bar{X}}{\bar{X^2} - \bar{X}^2}$$

$$\hat{\alpha}_m = \frac{\bar{X}^2}{\bar{X^2} - \bar{X}^2}$$

6) $X_1, \dots, X_n$

$$L(\alpha, \theta) = \frac{B^{n\alpha}}{[\Gamma(\alpha)]^n} (\prod x_i)^{\alpha-1} \exp\{-B \sum x_i\}$$

$$l(\alpha, \theta) = n\alpha \ln(B) - n \ln(\Gamma(\alpha)) + (\alpha-1) \sum \ln(x_i) - B \sum x_i$$

$$\textcircled{1} \frac{\partial l}{\partial \alpha} = n \ln(B) - n \left(\frac{\Gamma'(\alpha)}{\Gamma(\alpha)} \right) + \sum \ln(x_i) = 0$$

$$\textcircled{2} \frac{\partial l}{\partial B} = \frac{n\alpha}{B} - \sum x_i = 0$$

$$\frac{\partial^2 l}{\partial B^2} = -\frac{n}{B^2}$$

es invertierbar

$$\frac{\partial^2 l}{\partial \alpha^2} = -n \psi'(\alpha), \quad \frac{\partial^2 l}{\partial B^2} = -\frac{n\alpha}{B^2}$$

$$I_{\alpha\beta} = \begin{bmatrix} -\psi'(\alpha) & 1/B \\ 1/B & -\frac{\alpha}{B^2} \end{bmatrix}$$

$$\psi'(\alpha) \frac{\alpha}{B} - \frac{1}{B^2} \stackrel{!}{=} 0 \quad \psi'(\alpha) \alpha B = 1$$